I.1 — Iitaka Fibration

Fibrations in Birational Geometry Reading Notes • Summer 2026 Yi Li

Big picture

These notes give a first introduction to the **Iitaka fibration**. The guiding point is that a variety with non-negative Kodaira dimension carries a canonical rational map whose base has dimension $\kappa(X)$ and whose very general fibers have Kodaira dimension zero. As we will see, the Iitaka fibration is one of the building blocks of the minimal model program. It is also very useful when dealing with problems of intermediate Kodaira dimension, and in particular in the "divide and rule" construction: it isolates the positive Kodaira dimension on the base and leaves a Kodaira-dimension-zero problem on the general fiber.

Contents

1 Introduction

Let X be a normal complex variety, and suppose for the moment that $\kappa(X) \geq 0$. The pluricanonical systems $|mK_X|$ define rational maps

$$\Phi_{|mK_X|} \colon X \dashrightarrow \mathbb{P}H^0(X, mK_X)^\vee.$$

For m sufficiently large and divisible, these maps stabilize up to birational equivalence. The resulting birational fiber space is the *Iitaka fibration* of X:

$$X' \longrightarrow Y, \qquad \dim Y = \kappa(X), \qquad \kappa(F) = 0$$

for a very general fiber F.

The Iitaka fibration serves as a basic building block in the minimal model program, which suggests the following two possible outputs. If $\kappa(X) < 0$, the expected output is a Mori fiber space. If $\kappa(X) \geq 0$, the expected output is a minimal model whose canonical divisor is semiample, and hence induces the Iitaka fibration onto the canonical model Y:

$$\begin{tikzcd} & W \arrow[dl] \arrow[dr] & \\ X \arrow[rr, dashed] & & X_{\min} \arrow[d, "g"] \\ & & Y. \end{tikzcd}$$

In the minimal model program, $K_{X_{\min}} \sim_{\mathbb{Q}} g^*A$ for an ample $\mathbb{Q}$-divisor A on Y, and the general fiber has Kodaira dimension 0.

2 Kodaira Maps and Iitaka–Kodaira Dimension

2.1 Kodaira maps

Definition 2.1 (Kodaira map). Let X be a normal complex variety and let D be a **Cartier divisor** on X with $H^0(X, \mathcal{O}_X(D)) \neq 0$. The complete linear system $|D|$ defines a rational map

$$\Phi_{|D|} \colon X \dashrightarrow \mathbb{P}H^0(X, \mathcal{O}_X(D))^\vee.$$

If $s_0, \dots, s_N$ is a basis of $H^0(X, \mathcal{O}_X(D))$, then away from the base locus this map is

$$x \longmapsto [s_0(x) : \cdots : s_N(x)].$$

Equivalently, x is sent to the hyperplane of sections vanishing at x.

Remark 2.2. The map is undefined exactly at the base locus $\mathrm{Bs}\,|D|$, where every section of $\mathcal{O}_X(D)$ vanishes (though it is sometimes possible to extend the map across the indeterminacy locus). After replacing X by a resolution of the graph of $\Phi_{|D|}$, the Kodaira map becomes a morphism. This replacement is harmless for birational questions such as Kodaira dimension and the Iitaka fibration.

Definition 2.3 (Iitaka-Kodaira dimension of a divisor). Let L be a line bundle on a normal complex variety X. Set

$$\mathbb{N}(X, L) = \{m \in \mathbb{N} \mid H^0(X, L^{\otimes m}) \neq 0\},$$

which forms a semigroup. If $\mathbb{N}(X, L) = \varnothing$, define $\kappa(X, L) = -\infty$. Otherwise,

$$\kappa(X, L) = \max_{m \in \mathbb{N}(X,L)} \dim \Phi_{|mL|}(X),$$

where "image" means the closure of the image of the domain of definition; we will omit the closure in the rest of the notes. For a smooth complex variety X, the Kodaira dimension is $\kappa(X) = \kappa(X, K_X)$. For a singular variety, $\kappa(X)$ is defined via a smooth birational model, and one can show that this is independent of the choice of resolution. From the above definition, it is immediate that $\kappa(X, L) \leq \dim X$; when this upper bound is achieved, we call L big.

Example 2.4. Let X be a K3 surface and L a nef and big line bundle on X. For $m \gg 0$, the morphism $f := \varphi_{|mL|}$ contracts exactly the irreducible curves $C \subset X$ with $L \cdot C = 0$.
Negative definiteness of the contracted curves. Since L is nef and big, $L^2 > 0$, so by the Hodge index theorem the intersection form is negative definite on the hyperplane $L^\perp = \{D \in \mathrm{NS}(X) \otimes \mathbb{R} : D \cdot L = 0\}$: for $D \in L^\perp$ we have $D^2 \leq 0$, with equality only if $D \equiv 0$ numerically. Each contracted curve C_i satisfies $L \cdot C_i = 0$, so $C_i \in L^\perp$; since f contracts them to points, the classes $[C_i]$ are linearly independent. Hence for $D = \sum_i a_i C_i \neq 0$ we get $D^2 < 0$, that is, the intersection matrix $(C_i \cdot C_j)$ is negative definite.
Rationality of the curves. Let C be one such curve. Since $K_X \cong \mathcal{O}_X$, the genus formula gives $2p_a(C) - 2 = C^2$. As C is effective and nonzero it is not numerically trivial, so by the above $C^2 < 0$; since the K3 lattice is even and $p_a(C) \geq 0$, this forces $C^2 = -2$ and $p_a(C) = 0$. An irreducible curve of arithmetic genus 0 is smooth, so $C \cong \mathbb{P}^1$: each contracted curve is a smooth rational (-2)-curve, and the exceptional locus is a union of configurations of such curves.
ADE singularities. Any connected configuration of smooth rational curves with negative definite intersection matrix and all self-intersections -2 has dual graph given by a simply-laced Dynkin diagram of type A_n, D_n, E_6, E_7, or E_8 (Du Val); see Fig. 1.

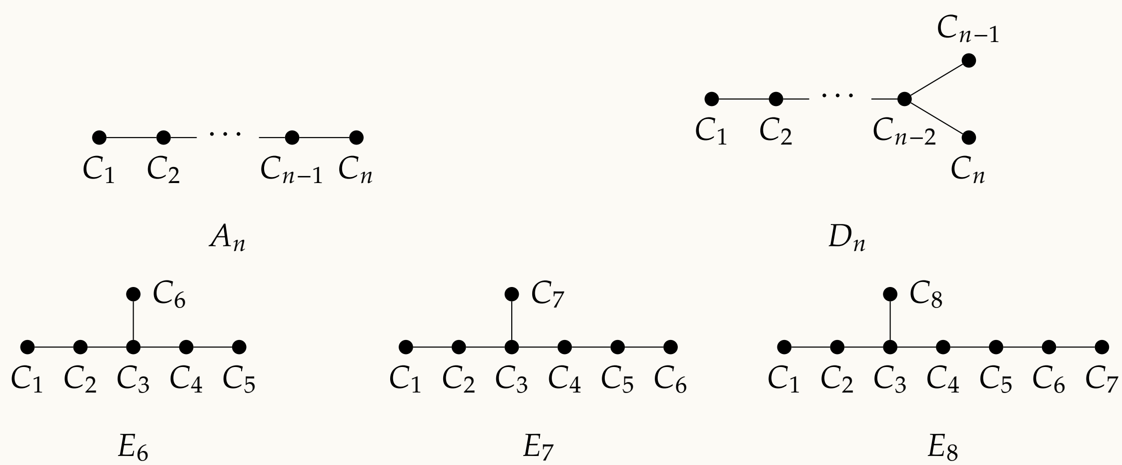

Figure 1: Dynkin diagrams / dual graphs of the exceptional curve configurations. Vertices are the (-2)-curves C_i; an edge joins C_i, C_j whenever $C_i \cdot C_j = 1$.

The Kodaira map preserves the K3 structure. Writing $K_X = f^*K_Y + \sum_i a_i C_i$ and intersecting with C_j gives, since $K_X \cdot C_j = 0$ and $(f^*K_Y) \cdot C_j = 0$,

$$0 = \sum_i a_i (C_i \cdot C_j), \qquad j = 1, \ldots, r.$$

Negative definiteness of $(C_i \cdot C_j)$ forces $a_i = 0$ for all i, so f is crepant: $K_X = f^*K_Y$. Thus $\omega_Y \cong \mathcal{O}_Y$, and Y is a K3 surface with ADE (Du Val) singularities along $f\left(\bigcup_i C_i\right)$.

The images of the Kodaira maps birationally stabilize; let us denote the image of the Kodaira map by $W_m := \Phi_{|mD|}(X)$.

Theorem 2.5 ([Uen75, Lemma 5.6])**.** *There exists a positive integer m_0 such that, for every integer $m \geq m_0$ with $m \in \mathbb{N}(D, V)$, we have*

$$\mathbb{C}(W_m) = \mathbb{C}\left(W_{m_0}\right).$$

That is, W_m is birationally equivalent to W_{m_0} for every $m \geq m_0$ with $m \in \mathbb{N}(D, V)$. In particular, $\kappa(V, D) = \dim W_{m_0}$.

Proof. Let $d > 0$ be the greatest common divisor of the integers belonging to $\mathbb{N}(D, V)$. Since $\mathbb{N}(D, V)$ is a semigroup, there exists a positive integer ℓ_0 such that $\ell d \in \mathbb{N}(D, V)$ for every $\ell \geq \ell_0$. Set $k :\,= \ell_0 d$, so that $nk \in \mathbb{N}(D, V)$ for every $n \geq 1$.
Fix $n \geq 1$ and let $\{\varphi_0, \ldots, \varphi_N\}$ be a basis of $H^0(V, \mathcal{O}_V(nkD))$. The rational map

$$\Phi_{|nkD|} \colon V \dashrightarrow W_{nk} \subset \mathbb{P}^N, \qquad x \mapsto [\varphi_0(x) \colon \cdots \colon \varphi_N(x)],$$

identifies, on the affine chart $\{X_0 \neq 0\}$, the function field of W_{nk} with

$$\mathbb{C}(W_{nk}) = \mathbb{C}(X_1/X_0, \ldots, X_N/X_0),$$

where we mean the restriction of the meromorphic function X_i/X_0 to W_{nk}. Pulling back by $\Phi_{|nkD|}$ gives an embedding

$$\Phi^*_{|nkD|} \colon \mathbb{C}(W_{nk}) \hookrightarrow \mathbb{C}(V), \qquad X_i/X_0 \mapsto g_i :\,= \varphi_i/\varphi_0,$$

so that $\mathbb{C}(W_{nk})$ is identified with the subfield $\mathbb{C}(g_1, \ldots, g_N) \subset \mathbb{C}(V)$.
Now fix a nonzero element $\psi \in H^0(V, \mathcal{O}_V(kD))$, which exists since $k \in \mathbb{N}(D, V)$. The products $\psi\varphi_0, \ldots, \psi\varphi_N$ lie in $H^0(V, \mathcal{O}_V((n+1)kD))$, and

$$\frac{\psi\varphi_i}{\psi\varphi_0} = \frac{\varphi_i}{\varphi_0} = g_i$$

for each i. Hence each g_i, viewed as an element of $\mathbb{C}(V)$, also lies in the (image of the) function field $\mathbb{C}(W_{(n+1)k})$. This yields an increasing chain of subfields of $\mathbb{C}(V)$,

$$\mathbb{C}(W_k) \subset \mathbb{C}(W_{2k}) \subset \mathbb{C}(W_{3k}) \subset \cdots \subset \mathbb{C}(V).$$

By the Siegel–Remmert–Thimm theorem, the field $\mathbb{C}(V)$ of meromorphic functions on the compact complex manifold V has finite transcendence degree $\operatorname{tr.deg}_{\mathbb{C}} \mathbb{C}(V) < +\infty$ over $\mathbb{C}$ and is finitely generated.
Set

$$L := \bigcup_{s \geq 1} \mathbb{C}(W_{sk}) \subseteq \mathbb{C}(V).$$

Then L is finitely generated over $\mathbb{C}$, say $L = \mathbb{C}(h_1, \ldots, h_m)$ with $h_1, \ldots, h_m \in L$. Since the chain $\{\mathbb{C}(W_{sk})\}_{s \geq 1}$ is increasing, there exists n_0 such that $h_1, \ldots, h_m \in \mathbb{C}(W_{n_0 k})$, whence $L \subseteq \mathbb{C}(W_{n_0 k})$. Combined with the trivial inclusion $\mathbb{C}(W_{n_0 k}) \subseteq L$, this gives

$$L = \mathbb{C}(W_{n_0 k}).$$

Since the chain is increasing, for every $s \geq n_0$ we have

$$\mathbb{C}(W_{n_0 k}) \subseteq \mathbb{C}(W_{sk}) \subseteq L = \mathbb{C}(W_{n_0 k}),$$

so that $\mathbb{C}(W_{sk}) = \mathbb{C}(W_{n_0 k})$ for all $s \geq n_0$.
Set $m_0 := (n_0 + 1)k$, and let $m \in \mathbb{N}(D, V)$ with $m \geq m_0$. Write

$$m = n_0 k + m',$$

so $m' \geq k$. Since $d \mid k$ and $d \mid m$, we have $d \mid m'$; as $m'/d \geq k/d = \ell_0$, it follows that $m' \in \mathbb{N}(D, V)$.
Fix a nonzero element $\psi' \in H^0(V, \mathcal{O}_V(m'D))$, and let $\varphi_0, \ldots, \varphi_N$ be a basis of $H^0(V, \mathcal{O}_V(n_0 kD))$ as before, so that the $g_i = \varphi_i / \varphi_0$ generate $\mathbb{C}(W_{n_0 k})$. The products $\psi'\varphi_0, \ldots, \psi'\varphi_N$ lie in $H^0(V, \mathcal{O}_V((n_0 k + m')D)) = H^0(V, \mathcal{O}_V(mD))$, and on the chart $\{Y_0 \neq 0\}$ of W_m we have

$$\Phi^*_{|mD|}\left(\frac{Y_i}{Y_0}\right) = \frac{\psi'\varphi_i}{\psi'\varphi_0} = \frac{\varphi_i}{\varphi_0} = g_i.$$

Hence each generator g_i of $\mathbb{C}(W_{n_0 k})$ lies in $\mathbb{C}(W_m)$, and therefore

$$\mathbb{C}(W_{n_0 k}) \subseteq \mathbb{C}(W_m). \tag{$*$}$$

For the reverse inclusion, since $m \in \mathbb{N}(D, V)$ the same multiplication argument as before (multiplying a basis of $H^0(V, \mathcal{O}_V(mD))$ by a nonzero section of $H^0(V, \mathcal{O}_V((mk - m)D))$, noting $d \mid mk - m$ and $mk - m$ large enough to lie in $\mathbb{N}(D, V)$) gives

$$\mathbb{C}(W_m) \subseteq \mathbb{C}(W_{mk}).$$

Since $m \geq m_0 = (n_0+1)k > n_0$, the integer mk is of the form sk with $s = m \geq n_0$, so

$$\mathbb{C}(W_{mk}) = \mathbb{C}(W_{n_0 k}). \tag{**}$$

Combining $(*)$ and $(**)$,

$$\mathbb{C}(W_{n_0 k}) \subseteq \mathbb{C}(W_m) \subseteq \mathbb{C}(W_{mk}) = \mathbb{C}(W_{n_0 k}),$$

which forces

$$\mathbb{C}(W_m) = \mathbb{C}(W_{n_0 k})$$

for every $m \in \mathbb{N}(D,V)$ with $m \geq m_0$. This completes the proof. ■

3 The Iitaka Fibration

We next introduce the Iitaka fibration, which can be viewed as the limit of the Kodaira maps.

3.1 Existence of Iitaka fibrations

Theorem 3.1 (Iitaka fibration theorem, [Uen75, Theorem 5.10])**.** *Let D be a Cartier divisor on a variety V with $\kappa(D,V) > 0$. Then there exist a complex manifold V^*, a complex manifold W^*, and a surjective morphism $f: V^* \to W^*$ with the following properties.*

(1) *There exists a modification $\pi: V^* \to V$.*

(2) $\dim W^* = \kappa(D,V)$.

(3) *For w in a dense (not necessarily open!) subset U of W^*, in the complex topology, the fibre $V^*_w = f^{-1}(w)$ is irreducible and non-singular.*

(4) $\kappa(\pi^* D_w, V^*_w) = 0$ *for every $w \in U$, where $\pi^* D_w$ denotes the restriction of $\pi^* D$ to V^*_w.*

(5) *The fibre space $f: V^* \to W^*$ is bimeromorphically equivalent to the fibre space associated with the meromorphic map $\Phi_{mD}: V \dashrightarrow W_m$, for every integer $m \geq m_0$ with $m \in \mathbb{N}(D,V)$.*

We may assume that V is normal, and, since $\kappa(D,V) > 0$, that D is effective.

Proof. Step 1: Resolving the Kodaira map. By Theorem 2.5 there exists $m_0 \in \mathbb{N}$ such that, for every $m \geq m_0$ with $m \in \mathbb{N}(D,V)$ and $W_m := \Phi_{mD}(V)$, one has $\dim W_m = \kappa(D,V)$ and $\mathbb{C}(W_m)$ is algebraically closed in $\mathbb{C}(V)$. Fix such an m and write $W := W_m$.
Let V^* be a non-singular model of the graph G of $\Phi_{mD}: V \dashrightarrow W$, and let $b: V^* \to V$ be the composite of the projection $G \to V$ with the desingularization $V^* \to G$; let $f: V^* \to W$

be the induced morphism, so that

$$\begin{tikzcd} V^* \arrow[d, "b"'] \arrow[dr, "f"] & \\ V \arrow[r, dashed, "\Phi_{mD}"'] & W. \end{tikzcd}$$

Since b is a modification and V^*, V are normal, the pullback gives an isomorphism

$$b^*\colon H^0(V, \mathcal{O}_V(\ell D)) \xrightarrow{\sim} H^0(V^*, \mathcal{O}_{V^*}(\ell D^*))$$

for every $\ell > 0$, where $D^* = b^*D$; consequently f coincides with Φ_{mD^*}. We may therefore, from the outset, replace V by V^*: that is, we may assume that $\Phi_{mD} = f$ is a morphism and V is non-singular. Write $L := \mathcal{O}_V(mD)$. As will become clear, $f\colon V \to W$ is the Iitaka fibration.

Step 2: The relative Kodaira map over W. Set $N + 1 = \dim_{\mathbb{C}} H^0(V, L)$, so that $W \subset \mathbb{P}^N$. A hyperplane $\lambda_0 X_0 + \cdots + \lambda_N X_N = 0$ cuts out a divisor H_λ on W, corresponding to a divisor $E_\lambda \in |mD|$. Let F be the fixed part of $|mD|$, put $E^*_\lambda = f^*H_\lambda$, so that $E_\lambda = E^*_\lambda + F$, and set $W_\lambda := W - H_\lambda$.

Since f is proper, $f_*(L^{\otimes n})$ is coherent on W for every $n > 0$; since W is projective, GAGA produces an algebraic coherent sheaf $\mathcal{F}_n$ on W^s (the associated scheme; we will omit this superscript without causing confusion) with $\mathcal{F}_n^{\mathrm{an}} \cong f_*(L^{\otimes n})$.

Note that W_λ is affine, as it is a closed subscheme of $\mathbb{P}^N \setminus H_\lambda$. By Cartan's Theorem A, $H^0(W_\lambda, \mathcal{F}_n)$ is generated, as an $H^0(W_\lambda, \mathcal{O}_{W_\lambda})$-module, by finitely many sections $\psi_0, \ldots, \psi_M$, as elements of

$$H^0(W_\lambda, f_*(L^{\otimes n})) = H^0(f^{-1}(W_\lambda), L^{\otimes n}).$$

Since

$$H^0(W_\lambda, \mathcal{F}_n) = H^0(W - H_\lambda, \mathcal{F}_n) \subset \bigcup_{e \geq 1} H^0(W, \mathcal{F}_n(eH_\lambda)),$$

and $\mathcal{F}_n(eH_\lambda) \cong \mathcal{F}_n \otimes \mathcal{O}_W([eH_\lambda])$ is coherent, we may choose e large enough that $\psi_i \in H^0(W, \mathcal{F}_n(enH_\lambda))$ for all i. By GAGA again,

$$H^0(W, \mathcal{F}_n(enH_\lambda)) \cong H^0(W, f_*(L^{\otimes n})(enH_\lambda)) = H^0(V, L^{\otimes n}(enE^*_\lambda)) \subset H^0(V, L^{\otimes n}(enE_\lambda)),$$

so each ψ_i may be regarded as a section in $H^0(V, L^{\otimes n}(enE_\lambda))$. Fixing $\eta \in H^0(V, \mathcal{O}_V(enE_\lambda))$ with $\mathrm{div}(\eta) = enE_\lambda$, we obtain

$$\eta\psi_0, \ldots, \eta\psi_M \ \in\ H^0(V, L^{\otimes(e+1)n}).$$

Let $\{\varphi_0, \ldots, \varphi_N\}$ be a basis of $H^0(V, L)$, and let $\psi_{M+1}, \ldots, \psi_{M'}$ be all monomials of degree n in the φ_j; then $\eta\psi_i \in H^0(V, L^{\otimes(e+1)n})$ for $i = M+1, \ldots, M'$ as well. These sections define a meromorphic map

$$h^{(n)}\colon V \dashrightarrow \mathbb{P}^{M'},$$

whose image we denote V_n. Since $\eta\psi_i \in H^0(V, L^{\otimes(e+1)n})$ for all $i = 0, \dots, M'$, and since $\{\varphi_j\}$ is a basis of $H^0(V, L)$, we get inclusions of function fields

$$\mathbb{C}(W) \subset \mathbb{C}(V_n) \subset \mathbb{C}(W_{(e+1)nm}),$$

where $W_{(e+1)nm}$ is the image of $\Phi_{(e+1)nmD}$. The first inclusion holds because the degree-n Veronese embedding $\nu_n \colon \mathbb{P}^N \hookrightarrow \mathbb{P}^{M'-M-1}$ is simply the map sending each coordinate to its degree-n monomial. Hence, if we focus on the last $M' - M$ coordinates of $h^{(n)}$, this is the composition $V \overset{\Phi_{mD}}{\dashrightarrow} W \subset \mathbb{P}^N \hookrightarrow \mathbb{P}^{M'-M-1}$. Hence W is defined using a subcollection of coordinates, so $\mathbb{C}(W) \subset \mathbb{C}(V_n)$.
Since W already realizes the Kodaira dimension, $\mathbb{C}(W_{(e+1)nm}) = \mathbb{C}(W)$, so all three fields coincide. Hence there is a commutative diagram

$$\begin{tikzcd} V \arrow[rr, dashed, "h^{(n)}"] \arrow[dr, "f"'] & & V_n \arrow[dl, dashed, "g_n"] \\ & W & \end{tikzcd}$$

with g_n birational. We will see in the next step that $V \to W$ is the model realizing the Iitaka fibration.

Step 3: Kodaira dimension zero on a very general fibre. Since $f_*(L^{\otimes n})$ is coherent, there is a proper algebraic subset $S_n \subset W$ such that $f_*(L^{\otimes n})|_{W-S_n}$ is locally free, L is flat over $W - S_n$, and base change holds:

$$f_*(L^{\otimes n})(w) \xrightarrow{\sim} H^0(V_w, L^{\otimes n}_{V_w}), \qquad w \in W - S_n.$$

For $w \in W_\lambda - S_n$, this identifies the restriction $\psi_{i,w}$ of ψ_i with an element of $H^0(V_w, L_w^{\otimes n})$. By generic smoothness, there is an algebraic subset $T \subset W$ such that $W - T$ is non-singular and every fibre V_w, $w \in W - T$, is non-singular. Since W_λ is affine, $\mathcal{F}_n$ is globally generated there, so $H^0(V_w, L_w^{\otimes n})$ is spanned by $\psi_{i,w}$, $i = 0, \dots, M$, for every $w \in W_\lambda - S_n$ via the restriction map

$$H^0(V, L^{\otimes n}) \to H^0(V_w, L^{\otimes n}_{V_w}).$$

Consequently, for $w \in W_\lambda - (S_n \cup T)$ the restriction $h_w^{(n)} \colon V_w \dashrightarrow \mathbb{P}^{M'}$ of $h^{(n)}$ is bimeromorphically equivalent to Φ_{mnD_w}, with image $h_w^{(n)}(V_w) = V_{n,w} = g_n^{-1}(w)$.
Since g_n is birational (as proved in the previous step), there are nowhere dense algebraic subsets $B_n \subset V_n$ and $A_n \subset W$ such that g_n restricts to an isomorphism $V_n - B_n \xrightarrow{\sim} W - A_n$. Hence, for every $w \in W_\lambda - (S_n \cup T \cup A_n)$, the fibre $V_{n,w} = g_n^{-1}(w)$ is a single point, so that $h_w^{(n)}(V_w)$ is a point and

$$\dim_{\mathbb{C}} H^0(V_w, \mathcal{O}_{V_w}(mnD_w)) = 1.$$

Set $Y := H_\lambda \cup T \cup \bigcup_{n\geq 1}(S_n \cup A_n)$, a countable union of nowhere dense algebraic subsets of W. Since W is a Baire space, $U := W - Y$ is dense in W. For every $w \in U$ and every $n \geq 1$,

$$\dim_{\mathbb{C}} H^0(V_w, \mathcal{O}_{V_w}(nmD_w)) = 1.$$

Since D is effective, $H^0(V_w, \mathcal{O}_{V_w}(nD_w)) \subset H^0(V_w, \mathcal{O}_{V_w}(nmD_w))$ for every n, so $\dim_{\mathbb{C}} H^0(V_w, \mathcal{O}_{V_w}(nD_w)) \leq 1$, and therefore

$$\kappa(D_w, V_w) = 0 \qquad \text{for every } w \in U.$$

Step 4: Desingularizing the base. Let $\delta\colon W^* \to W$ be a desingularization of W, obtained as a finite succession of blow ups. Then $g := \delta^{-1} \circ f\colon V \dashrightarrow W^*$ is a generically surjective meromorphic map, holomorphic on $V' := V - f^{-1}(T)$, since $W - T$ is non-singular.
Let G be the graph of g and let $\pi^*\colon V^* \to G$ be a desingularization of G; set $\pi := p_V \circ \pi^*\colon V^* \to V$. Then π is a modification inducing an isomorphism $\pi^{-1}(V') \xrightarrow{\sim} V'$, and, since $W - T$ and $W^* - \delta^{-1}(T)$ are isomorphic, the induced morphism $f^* := f \circ \pi\colon V^* \to W^*$ together with $U^* := \delta^{-1}(U)$ satisfies the properties of the theorem.

∎

3.2 Relative Iitaka Fibrations

We next provide a relative version of the Iitaka fibration theorem. Note that the proof requires a projectivity assumption; it is expected that the same result holds for proper morphisms, but unfortunately I do not know how to prove this.

Theorem 3.2 (Relative Iitaka fibration, [Nak04; BC15])**.** *Let $f\colon X \to Z$ be a projective morphism of normal quasi-projective varieties, and let D be a $\mathbb{Q}$-Cartier divisor on X. Let F be a general fiber of f and assume*

$$r := \kappa(D|_F) \geq 0.$$

Then, after replacing X by a resolution $h\colon W \to X$, there is a contraction $g\colon W \to S$ over Z such that, if V is a general fiber of $S \to Z$ and G is a general fiber of g, then

$$\dim V = r, \qquad \kappa(h^*D|_G) = 0.$$

$$\begin{tikzcd} W \arrow[r, "h"] \arrow[d, "g"'] & X \arrow[d, "f"] \\ S \arrow[r] & Z. \end{tikzcd}$$

Proof. The idea is due to Birkar–Chen. Choose a sufficiently ample divisor A on Z so that

$$H^0(X, \mathcal{O}_X(mD + f^*A)) \neq 0$$

for some divisible $m > 0$, and set $L = D + f^*A$. Fujita's Iitaka dimension identity (see [Iit82, Exercise 10.1]) gives

$$\kappa(L) = \kappa(D|_F) + \dim Z = r + \dim Z.$$

Take the ordinary Iitaka fibration of L on a resolution of X:

$$\varphi\colon W \longrightarrow S.$$

We show that this fibration factors over Z, so that we have the following diagram.

$$\begin{tikzcd} W \arrow[r, "h"] \arrow[d, "g"'] & X \arrow[d, "f"] \\ S \arrow[r] & Z. \end{tikzcd}$$

To see this, let $\psi = f \circ h\colon W \to Z$. For a sufficiently divisible multiple, the linear system defining φ has the form

$$|kmL| = |M| + F_{\text{fix}}, \qquad M \sim \varphi^* H$$

for an ample divisor H on S. Since $mD + f^*A$ has a nonzero section, after pullback to W we may write

$$mh^*D + \psi^*A \sim B \geq 0.$$

Thus

$$mh^*L \sim B + (m-1)\psi^*A.$$

For a large multiple k, subtracting the fixed part of $|kmh^*L|$ gives

$$M \sim N + k(m-1)\psi^*A$$

with $N \geq 0$. If C is a curve contracted by φ and chosen away from $\operatorname{Supp} N$, then

$$0 = M \cdot C = N \cdot C + k(m-1)\psi^*A \cdot C.$$

Both terms on the right are non-negative, so $\psi^*A \cdot C = 0$. Since A is ample, $\psi(C)$ is a point. Thus curves contracted by φ are contracted by ψ, and the rigidity lemma gives a morphism $S \to Z$.

Finally, by Fujita's addition formula

$$\dim S = \kappa(L) = r + \dim Z,$$

a general fiber V of $S \to Z$ has dimension r. On a general fiber G of g, the ordinary Iitaka theorem gives $\kappa(h^*L|_G) = 0$. Since G lies inside a fiber of ψ, the divisor ψ^*A restricts trivially to G. Hence $h^*L|_G = h^*D|_G$, and $\kappa(h^*D|_G) = 0$. ■

3.3 Universal properties of Iitaka fibrations

The Iitaka fibration has the following universal property.

Proposition 3.3 ([Nak87]). *Let*

$$g : X \dashrightarrow G/S$$

be a proper surjective meromorphic map. If
(a) $\dim G = \dim S + \kappa(X/S, D)$,
(b) there exist a proper bimeromorphic morphism $\gamma : V \longrightarrow X$ *over* S *such that* $g \circ \gamma : V \longrightarrow G$ *is a morphism with*

$$\kappa(V/G; \gamma^* D) = 0,$$

then g *is properly bimeromorphically equivalent to the Iitaka fibration (or equivalently the Kodaira map for* $m \gg 1$*).*

4 Several applications of Iitaka fibrations

We next introduce several applications of the Iitaka fibration in minimal model theory and abundance.

4.1 Good Minimal Models and Abundance

Theorem 4.1 (Abundance and good minimal model, [DHP13, Remark 2.6]). *For pseudo-effective klt pairs, the existence of good minimal models is equivalent to abundance:*

$$\kappa_\sigma(K_X + \Delta) = \kappa(K_X + \Delta).$$

Proof. Assume first that (X, Δ) has a good minimal model (X', Δ'). Then $K_{X'} + \Delta'$ is semiample, so it defines a morphism

$$f : X' \longrightarrow Z = \operatorname{Proj} R(X', K_{X'} + \Delta')$$

and

$$K_{X'} + \Delta' \sim_{\mathbb{Q}} f^* A$$

for an ample $\mathbb{Q}$-divisor A on Z. Since Kodaira dimension and numerical Kodaira dimension are birational invariants for this purpose, and since semiample divisors are abundant,

$$\kappa_\sigma(K_X + \Delta) = \kappa_\sigma(K_{X'} + \Delta') = \dim Z = \kappa(K_{X'} + \Delta') = \kappa(K_X + \Delta).$$

Conversely, assume abundance. If $\kappa_\sigma(K_X + \Delta) = \dim X$, then $K_X + \Delta$ is big and BCHM gives a good minimal model. If $\kappa_\sigma(K_X + \Delta) = 0$, the numerical-dimension-zero case gives a good minimal model (see e.g. [Gon11]). Thus we may assume

$$0 < \kappa(K_X + \Delta) = \kappa_\sigma(K_X + \Delta) < \dim X.$$

Let

$$f\colon X \longrightarrow Z = \operatorname{Proj} R(K_X + \Delta)$$

be a birational model of the Iitaka fibration of $K_X + \Delta$, and let F be a very general fiber. The Iitaka fibration gives

$$\dim Z = \kappa(K_X + \Delta).$$

By the addition formula for numerical Kodaira dimension and its converse in this setting,

$$\kappa_\sigma(K_X + \Delta) = \kappa_\sigma(K_F + \Delta|_F) + \dim Z.^{a}$$

Therefore

$$\kappa_\sigma(K_F + \Delta|_F) = 0.$$

The numerical-dimension-zero case gives a good minimal model for the very general fiber. Applying Lai's theorem ([Lai11]) — if the general fiber of the Iitaka fibration admits a good minimal model, then so does the total space — one obtains a good minimal model for (X, Δ). ∎

[a] Here are some details for this dimension equality. The easy addition formula proved above is not enough; we instead apply the addition formula of [Nak04, Theorem V.4.1]. As noted by Fujino, Nakayama's proof is incomplete, but fortunately we only need the first part, which states

$$\kappa_\sigma(D + f^*Q) \geq \kappa_\sigma(D, X/Y) + \kappa_\sigma(Q),$$

which can be proved using [Fuj17]. Indeed, for f we have $K_X + \Delta \sim_{\mathbb{Q}} f^*H + D$, where H is some ample divisor. Substituting into the above addition formula, we get

$$\kappa_\sigma(K_X + \Delta) \geq \kappa_\sigma(K_F + \Delta|_F) + \kappa_\sigma(H),$$

with $\kappa_\sigma(Q) = \dim Z$.

4.2 Easy Addition Formula

Theorem 4.2 (Easy addition, [Uen75]). *Let $f\colon V \longrightarrow W$ be a fiber space between nonsingular complex varieties, and let D be a Cartier divisor on V. Then there is a dense open subset $U \subseteq W$ such that for every $w \in U$,*

$$\kappa(D, V) \leq \dim W + \kappa(D_w, V_w), \qquad V_w = f^{-1}(w), \quad D_w = D|_{V_w}.$$

Proof. If $\kappa(D, V) = -\infty$, there is nothing to prove. Fix $m \in \mathbb{N}(D, V)$ and set

$$\mathcal{L}_m = f_*\mathcal{O}_V(mD).$$

By generic freeness and base change, there is a dense open subset $U^{(m)} \subseteq W$ such that $\mathcal{L}_m|_{U^{(m)}}$ is locally free, V_w is nonsingular, and

$$\mathcal{L}_m \otimes \mathbb{C}(w) \simeq H^0(V_w, \mathcal{O}_{V_w}(mD_w))$$

for every $w \in U^{(m)}$. Let
$$g^{(m)}\colon \mathbb{P}_{U^{(m)}}(\mathcal{L}_m|_{U^{(m)}}) \to U^{(m)}$$
be the associated projective bundle. The natural evaluation map defines a relative Kodaira map
$$h^{(m)}\colon f^{-1}(U^{(m)}) \dashrightarrow \mathbb{P}_{U^{(m)}}(\mathcal{L}_m|_{U^{(m)}})/U^{(m)}.$$
By the base change isomorphism above, the restriction
$$h_w^{(m)}\colon V_w \dashrightarrow \mathbb{P}(\mathcal{L}_m \otimes \mathbb{C}(w))$$
is exactly the Kodaira map $\Phi_{|mD_w|}$ for $w \in U^{(m)}$ (here we use that base change holds, i.e. $\mathcal{L}_m(w) \cong H^0(V_w, \mathcal{O}_{V_w}(mD_w))$ for $w \in U^{(m)}$).
Shrinking $U^{(m)}$ to some Zariski open subset U, we may assume that the dimension of the image of this fiberwise Kodaira map is constant, by upper semi-continuity of the fiber dimension. Then
$$\dim h^{(m)}(V) = \dim W + \dim \Phi_{|mD_w|}(V_w)$$
for w in the dense open subset $U \subset U^{(m)}$. On the other hand, we have the canonical isomorphism $H^0(W, \mathcal{L}_m) = H^0(V, \mathcal{O}_V(mD))$. Using a basis of $H^0(V, \mathcal{O}_V(mD))$ we can construct the rational map
$$h : \mathbb{P}\left(\mathcal{L}_m\right) \dashrightarrow \mathbb{P}^N, \quad N = \dim_{\mathbb{C}} H^0\left(V, \mathcal{O}_V(mD)\right),$$
whose composition with $h^{(m)}$ is the Kodaira map $\Phi_{|mD|}$. Indeed, we use the tautological line bundle on the projective bundle $\mathbb{P}(\mathcal{L}_m) \xrightarrow{g^{(m)}} W$. By the pushforward relation for the tautological line bundle, $g_*^{(m)}\mathcal{O}_{\mathbb{P}(\mathcal{L}_m)}(1) \cong \mathcal{L}_m$ (see [Har77, Proposition 7.11]), and hence
$$H^0\big(\mathbb{P}(\mathcal{L}_m), \mathcal{O}(1)\big) \;\cong\; H^0\big(W, g_*^{(m)}\mathcal{O}(1)\big) \;=\; H^0(W, \mathcal{L}_m).$$
On the other hand, since $\mathcal{L}_m = f_*\mathcal{O}_V(mD)$ and $f^{-1}(W) = V$, we have the canonical isomorphism
$$H^0(W, \mathcal{L}_m) \;\cong\; H^0\big(V, \mathcal{O}_V(mD)\big).$$
Composing these two isomorphisms identifies $H^0\big(\mathbb{P}(\mathcal{L}_m), \mathcal{O}(1)\big)$ with $H^0(V, \mathcal{O}_V(mD))$. In particular, the basis $s_0, \dots, s_N$ of $H^0(V, \mathcal{O}_V(mD))$ defining Φ_{mD} corresponds under this identification to a basis $t_0, \dots, t_N$ of $H^0\big(\mathbb{P}(\mathcal{L}_m), \mathcal{O}(1)\big)$. This basis (t_i) defines a rational map
$$h : \mathbb{P}(\mathcal{L}_m) \dashrightarrow \mathbb{P}^N, \qquad x \mapsto [t_0(x) : \cdots : t_N(x)].$$
On the other hand, the relative Kodaira map $h^{(m)} : V \dashrightarrow \mathbb{P}(\mathcal{L}_m)$ is by construction induced by the canonical surjection for the tautological line bundle,
$$f^*\mathcal{L}_m = f^* f_*\mathcal{O}_V(mD) \longrightarrow \mathcal{O}_V(mD),$$

so that, on the open locus where $h^{(m)}$ is defined (the locus where the above map is surjective, i.e. globally generated),

$$\mathcal{O}_V(mD) \;\cong\; (h^{(m)})^*\mathcal{O}_{\mathbb{P}(\mathcal{L}_m)}(1).$$

By naturality, this identification arises from the same adjunction morphism as the isomorphism above; consequently the pullback map on sections

$$(h^{(m)})^* : H^0\big(\mathbb{P}(\mathcal{L}_m), \mathcal{O}(1)\big) \longrightarrow H^0\big(V, \mathcal{O}_V(mD)\big)$$

coincides with the isomorphism constructed above. In particular $(h^{(m)})^*(t_i) = s_i$ for every i.
Combining the two constructions, we obtain

$$(h \circ h^{(m)})^*\mathcal{O}_{\mathbb{P}^N}(1) \;\cong\; \mathcal{O}_V(mD), \qquad (h \circ h^{(m)})^*(x_i) = (h^{(m)})^*(t_i) = s_i,$$

where $x_0, \ldots, x_N$ are the homogeneous coordinates on $\mathbb{P}^N$. That is, $h \circ h^{(m)}$ is precisely the map $x \mapsto [s_0(x) : \cdots : s_N(x)]$ with respect to the chosen basis of $H^0(V, \mathcal{O}_V(mD))$. Since Φ_{mD} is by definition the rational map associated to this basis, we conclude

$$h \circ h^{(m)} = \Phi_{mD}$$

wherever both sides are defined.
Hence the image of $\Phi_{|mD|}$ is dominated by $h^{(m)}(V)$, so

$$\dim \Phi_{|mD|}(V) \leq \dim h^{(m)}(V) = \dim W + \dim \Phi_{|mD_w|}(V_w).$$

Now choose m sufficiently large so that $\dim \Phi_{|mD|}(V) = \kappa(D, V)$. Since

$$\dim \Phi_{|mD_w|}(V_w) \leq \kappa(D_w, V_w),$$

we obtain

$$\kappa(D, V) \leq \dim W + \kappa(D_w, V_w)$$

for w in a dense open subset of W. ■

4.3 Finite Generation of the Canonical Ring

The Iitaka fibration can reduce the finite generation problem to the general type case (the proof of this result first appeared in Fujino–Mori's paper [FM00]). The proof below is written in the form suggested by Fujino's relative and analytic version [Fuj22]. The idea of the proof, however, is essentially the same as the original proof [FM00].

Theorem 4.3 (Finite generation of the canonical ring, [Fuj22, Theorem 1.18]). *Let (X, Δ) be a klt pair, where $K_X + \Delta$ is $\mathbb{Q}$-Cartier, and let $\pi : X \longrightarrow Y$ be a projective morphism of complex analytic spaces. Then*

$$R\,(X/Y, K_X + \Delta) = \bigoplus_{m \in \mathbb{N}} \pi_* \mathcal{O}_X\,(\lfloor m\,(K_X + \Delta) \rfloor)$$

is a locally finitely generated graded $\mathcal{O}_Y$-algebra.

Proof. For simplicity, we present the proof in the absolute setting; for the relative setting the argument is essentially the same, except that one needs to use the relative Iitaka fibration. Note that the big case is the finite generation theorem of BCHM [BCHM10]. After taking a log resolution and replacing by a suitable birational model, assume that X is smooth and $\operatorname{Supp} \Delta$ has simple normal crossings.
Let

$$f : X \dashrightarrow Z$$

be the Iitaka fibration of $K_X + \Delta$. After resolving the indeterminacy, we may assume that f is a morphism with connected fibers and that Z is smooth. The defining property of the Iitaka fibration is

$$\dim Z = \kappa(K_X + \Delta), \qquad \kappa((K_X + \Delta)|_F) = 0$$

for a very general fiber F. By [Fuj22], after taking some base change and replacing Z by a suitable birational model we may assume that

$$K_X + \Delta \sim_{\mathbb{Q}} f^*(K_Z + B_Z + M_Z),$$

where B_Z is the discriminant part and M_Z is the moduli part. Consequently, after taking a sufficiently divisible Veronese subring, the canonical ring of (X, Δ) is identified with

$$\bigoplus_{m \geq 0} H^0(Z, \mathcal{O}_Z(mk(K_Z + B_Z + M_Z)))$$

for some $k > 0$.
This is the point where the Iitaka fibration is essential. Since Z is the base of the Iitaka fibration,

$$\kappa(K_Z + B_Z + M_Z) = \dim Z.$$

Thus $K_Z + B_Z + M_Z$ is big. In other words, the Iitaka fibration has converted the original non-big divisor $K_X + \Delta$ on X into a big divisor on its canonical base.
Since M_Z is nef and $K_Z + B_Z + M_Z$ is big, write

$$K_Z + B_Z + M_Z \sim_{\mathbb{Q}} A + G$$

with A ample and $G \geq 0$. For $0 < \varepsilon \ll 1$, the pair

$$(Z, \Delta_Z), \qquad \Delta_Z = B_Z + \varepsilon G + H$$

is klt for a general small effective $\mathbb{Q}$-divisor $H \sim_{\mathbb{Q}} M_Z + \varepsilon A$, and

$$K_Z + \Delta_Z \sim_{\mathbb{Q}} (1+\varepsilon)(K_Z + B_Z + M_Z).$$

By the truncation principle for finite generation, it is enough to prove finite generation for

$$R(Z, K_Z + \Delta_Z).$$

But $K_Z + \Delta_Z$ is big and (Z, Δ_Z) is klt, so BCHM applies. Hence $R(Z, K_Z + \Delta_Z)$ is finitely generated, and therefore so is $R(X, K_X + \Delta)$. ■

5 Further Discussion

We end these notes with some further topics about the Iitaka fibration, leaving the details to interested readers.

Remark 5.1 (Iitaka fibration fibered with good minimal model)**.** Lai [Lai11] shows that, for terminal projective varieties, the existence of good minimal models for the general fibers of the Iitaka fibration implies the existence of a good minimal model for the total space. This gives a standard strategy for reducing a problem of intermediate Kodaira dimension to the case of Kodaira dimension zero.

Remark 5.2 (Classification of "non-positively curved" varieties)**.** Varieties with non-positive curvature share many rigidity properties. Höring [Hör13] proved that for a compact Kähler manifold X with Ω^1_X nef and K_X semi-ample, after an étale base change the Iitaka fibration $X \to Y$ becomes a smooth fibration whose fibers are all complex tori, with Y a projective manifold satisfying K_Y ample. It is conjectured that the same result holds without the assumption that K_X is semi-ample.

Remark 5.3 (Effective Iitaka fibration conjecture and boundedness)**.** One of the central problems in boundedness is the Effective Iitaka fibration conjecture. In the proof of the Iitaka fibration theorem, we showed that the maps $\Phi_{|mK_X|}$ eventually stabilize. The effective Iitaka fibration and boundedness problems ask, further, for a uniform bound on such an m; see Birkar–Zhang [BZ16].

Remark 5.4 (Iitaka fibration in the abundance conjecture)**.** The Iitaka fibration is also a useful tool in the proof of the abundance conjecture. Lazić–Peternell [LP20] use the Iitaka fibration machine for an induction on dimension: given data (X, Δ, L) admitting a non-trivial relative Iitaka fibration $X \to T/Z$, they descend the data via the Iitaka fibration to (T, Δ_T, L_T), with $\dim T < \dim X$, so that generalized abundance for (T, Δ_T, L_T), together with the pulled-back relation $K_X + \Delta + L \equiv \tau^*(K_T + \Delta_T + L_T)$, implies generalized abundance for the original data.

6 Appendix. Abundance conjecture when $\kappa(X) \geq \dim X - 3$

In this appendix, we present the proof of the abundance conjecture when $\kappa(X) \geq \dim X - 3$ using Iitaka fibration reduction. This result will also be used in the next lecture on the Albanese mapping; the idea of the proof first appeared in Kawamata's paper [Kaw85].

Theorem 6.1 ([Kaw85, Theorem 7.3])**.** *Let (X, Δ) be a projective klt pair, and assume that $K_X + \Delta$ is nef and $\kappa(K_X + \Delta) > 0$. If the minimal model conjecture and the abundance conjecture hold in dimension* $\dim X - \kappa(K_X + \Delta)$*, then*

$$\kappa(X, K_X + \Delta) = \nu(X, K_X + \Delta).$$

In particular, since the minimal model and abundance conjectures hold in dimension ≤ 3, the abundance conjecture holds whenever $\kappa(K_X + \Delta) \geq \dim X - 3$.

Proof. The key tool is the Iitaka fibration. First, by taking a terminal modification, we may assume that the pair (X, Δ) is terminal. Since $\kappa(K_X + \Delta) > 0$, we may take the Iitaka fibration on a smooth model, as in the following diagram:

$$\begin{tikzcd} Y \arrow[r, "\mu"] \arrow[d, "f"'] & X \\ Z & \end{tikzcd}$$

we can assume that Y and Z are smooth. By the defining property of the Iitaka fibration we have

$$\kappa(K_W + \Delta_W) = 0,$$

where W denotes a general fiber. Since the general fiber has dimension $\dim X - \dim Z = \dim X - \kappa(K_X + \Delta)$, the minimal model and abundance conjectures hold for it, so we can run the minimal model program

$$\sigma : W \dashrightarrow W_{\min}.$$

Since $\kappa(K_{W_{\min}} + \Delta_{W_{\min}}) = \nu(K_{W_{\min}} + \Delta_{W_{\min}}) = 0$, we have $K_{W_{\min}} + \Delta_{W_{\min}} \sim_{\mathbb{Q}} 0$. By blowing up Y we may assume that $W \dashrightarrow W_{\min}$ is a morphism. Then, pulling back,

$$K_Y + \Delta_Y \sim_{\mathbb{Q}} \mu^*(K_X + \Delta) + \sum_i r_i F_i, \qquad K_W + \Delta_W \sim_{\mathbb{Q}} \sum_j s_j G_j$$

with $r_i, s_j \geq 0$, where F_i, G_j are exceptional for μ, σ respectively. Hence, after adjunction on a general fiber, we get

$$\mu^*(K_X + \Delta)|_W \sim_{\mathbb{Q}} \sum_j s_j G_j - \sum_i r_i F_i|_W.$$

We show that the terms on the right-hand side vanish, using the Hodge index theorem and the negativity lemma.
We finish the proof by producing ample divisors L_1, L_2 on Z and a member $D \in |m\mu^*K_X + f^*L_1|$, for some $m > 0$, with $0 \leq D \leq f^*L_2$; for simplicity we set $N = \mu^*K_X$. Indeed, the linear system $|m\mu^*K_X + f^*L_1|$ is non-empty: for a sufficiently ample L_1 we have $h^0(Y, m\mu^*K_X + f^*L_1) > 0$. Next we check that the member D is always vertical. Suppose, for contradiction, that D is horizontal; then its restriction to the general fiber W is an effective $\mathbb{R}$-divisor, so $H'^{n-1} \cdot D|_W > 0$ for an ample divisor H'. However, $D \sim mN + f^*L_1$, and restricting to W and using $N|_W \equiv 0$ (shown by induction above), together with $f^*L_1|_W \equiv 0$, gives $D|_W \equiv 0$, a contradiction. Hence $D = \sum_i a_i D_i$ is vertical, i.e. each $P_i := f(D_i)$ is a proper subvariety of Z. Choosing ample divisors H_i through P_i and passing to a sufficiently large multiple, we obtain an ample divisor $L_2 = \sum_i H_i$ with $0 \leq D \leq f^*L_2$.
We next claim that

$$\nu(X) = \nu(Y, \mu^*K_X) \leq \dim Z = \kappa(X),$$

where the reverse inequality holds trivially. We first observe that $D \leq f^*L_2$ gives

$$f^*L_2 - mN \geq 0,$$

and therefore

$$H^{n-k-1} \cdot ((f^*L_2)^{k+1} - (mN)^{k+1}) = \left(\sum_{j=0}^{k} (f^*L_2)^j (mN)^{k-j} (f^*L_2 - mN)\right) \cdot H^{n-k-1} \geq 0,$$

where $k = \dim Z$. On the other hand $L_2^{k+1} \equiv 0$, hence $\nu(N) \leq \dim Z$. ■

References

[BCHM10] C. Birkar et al. “Existence of minimal models for varieties of log general type”. In: *Journal of the American Mathematical Society* 23.2 (2010), pp. 405–468.

[BC15] Caucher Birkar and Jungkai Alfred Chen. “Varieties fibred over abelian varieties with fibres of log general type”. In: *Adv. Math.* 270 (2015), pp. 206–222. ISSN: 0001-8708,1090-2082. DOI: 10.1016/j.aim.2014.10.023. URL: https://doi.org/10.1016/j.aim.2014.10.023.

[BZ16] Caucher Birkar and De-Qi Zhang. “Effectivity of Iitaka fibrations and pluricanonical systems of polarized pairs”. In: *Publ. Math. Inst. Hautes Études Sci.* 123 (2016), pp. 283–331. ISSN: 0073-8301,1618-1913. DOI: 10.1007/s10240-016-0080-x. URL: https://doi.org/10.1007/s10240-016-0080-x.

[DHP13] Jean-Pierre Demailly, Christopher D. Hacon, and Mihai Păun. “Extension theorems, non-vanishing and the existence of good minimal models”. In: *Acta Math.* 210.2 (2013), pp. 203–259. ISSN: 0001-5962,1871-2509. DOI: 10.1007/s11511-013-0094-x. URL: https://doi.org/10.1007/s11511-013-0094-x.

[Fuj22] Osamu Fujino. *Minimal model program for projective morphisms between complex analytic spaces*. 2022. arXiv: 2201.11315 [math.AG]. URL: https://arxiv.org/abs/2201.11315.

[Fuj17] Osamu Fujino. “On subadditivity of the logarithmic Kodaira dimension”. In: *J. Math. Soc. Japan* 69.4 (2017), pp. 1565–1581. ISSN: 0025-5645,1881-1167. DOI: 10.2969/jmsj/06941565. URL: https://doi.org/10.2969/jmsj/06941565.

[FM00] Osamu Fujino and Shigefumi Mori. “A canonical bundle formula”. In: *J. Differential Geom.* 56.1 (2000), pp. 167–188. ISSN: 0022-040X,1945-743X. URL: http://projecteuclid.org/euclid.jdg/1090347529.

[Gon11] Yoshinori Gongyo. “On the minimal model theory for dlt pairs of numerical log Kodaira dimension zero”. In: *Math. Res. Lett.* 18.5 (2011), pp. 991–1000. ISSN: 1073-2780. DOI: 10.4310/MRL.2011.v18.n5.a16. URL: https://doi.org/10.4310/MRL.2011.v18.n5.a16.

[Har77] Robin Hartshorne. *Algebraic geometry*. Vol. No. 52. Graduate Texts in Mathematics. Springer-Verlag, New York-Heidelberg, 1977, pp. xvi+496. ISBN: 0-387-90244-9.

[Hör13] Andreas Höring. “Manifolds with nef cotangent bundle”. In: *Asian J. Math.* 17.3 (2013), pp. 561–568. ISSN: 1093-6106,1945-0036. DOI: 10.4310/AJM.2013.v17.n3.a7. URL: https://doi.org/10.4310/AJM.2013.v17.n3.a7.

[Iit82] Shigeru Iitaka. *Algebraic geometry*. Vol. 24. North-Holland Mathematical Library. An introduction to birational geometry of algebraic varieties, Graduate Texts in Mathematics, 76. Springer-Verlag, New York-Berlin, 1982, pp. x+357. ISBN: 0-387-90546-4.

[Kaw85] Y. Kawamata. “Pluricanonical systems on minimal algebraic varieties”. In: *Invent. Math.* 79.3 (1985), pp. 567–588. ISSN: 0020-9910,1432-1297. DOI: 10.1007/BF01388524. URL: https://doi.org/10.1007/BF01388524.

[Lai11] Ching-Jui Lai. “Varieties fibered by good minimal models”. In: *Math. Ann.* 350.3 (2011), pp. 533–547. ISSN: 0025-5831,1432-1807. DOI: 10.1007/s00208-010-0574-7. URL: https://doi.org/10.1007/s00208-010-0574-7.

[LP20] Vladimir Lazić and Thomas Peternell. "On generalised abundance, I". In: *Publ. Res. Inst. Math. Sci.* 56.2 (2020), pp. 353–389. ISSN: 0034-5318,1663-4926. DOI: 10.4171/prims/56-2-3. URL: https://doi.org/10.4171/prims/56-2-3.

[Nak87] Noboru Nakayama. "The lower semicontinuity of the plurigenera of complex varieties". In: *Algebraic geometry, Sendai, 1985*. Vol. 10. Adv. Stud. Pure Math. North-Holland, Amsterdam, 1987, pp. 551–590. ISBN: 0-444-70313-6. DOI: 10.2969/aspm/01010551. URL: https://doi.org/10.2969/aspm/01010551.

[Nak04] Noboru Nakayama. *Zariski-decomposition and abundance*. Vol. 14. MSJ Memoirs. Mathematical Society of Japan, Tokyo, 2004, pp. xiv+277. ISBN: 4-931469-31-0.

[Uen75] Kenji Ueno. *Classification theory of algebraic varieties and compact complex spaces*. Vol. Vol. 439. Lecture Notes in Mathematics. Notes written in collaboration with P. Cherenack. Springer-Verlag, Berlin-New York, 1975, pp. xix+278.